

DATA MANAGEMENT TECHNOLOGIES

UNIT 1 - TASK 2: PART 1

TASK

1. Identify the relations to represent Customer, Product and Sales including the attributes for each relation
2. Identify the candidate keys for each relation
3. Define your chosen Primary Key for the Customer, Product and Sales relations
4. Define what you see as appropriate Foreign Keys for each relation where you feel that is relevant
5. Identify any relationships and their cardinality

RESPONSE

The related database is a customer sales database, with information on customers, suppliers, and orders.

1. **The relations include these attributes:**

Customer relation: customer ID, name, contact details, credit provision

Product relation: product ID, name, storage conditions, shelf life, location

Sales relation: order ID, customer ID, credit provision, transaction date

2. **Candidate Keys**

Customer: customer ID, email address, phone number

Product: product ID, product name

Sales: order ID

3. **Primary keys:**

Customer: customer ID

Product: product ID

Sales: order ID

4. **Foreign keys:**

Customer: N/A (no foreign keys, each customer record is rather independent)

Product: supplier ID (references primary key of supplier table)

Sales: customer ID (references primary key of customer table)

5. **Relationships**

- Supplier to Products - **one to many**.
Each supplier can supply multiple products.
- Customers to Orders - **one to many**.
Each customer can have multiple orders
- Employee record to Pay record - **one to many**.
Each employee has multiple payroll records (one for each payday)
- Order to Products - **many to many**.
There are many products and many orders. Each product can appear in multiple orders.
- Order to Invoice number - **one to one**.
Each order has one invoice.